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Basic Reproduction Number and Epidemic Threshold

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Introduction

The basic reproduction number R_0 for the pair-approximation SIR on a homogeneous unclustered n -regular network is

$$R_0 = \frac{\tau(n-2)}{\gamma}.$$

The “ $n-2$ ” — rather than $n-1$ as in the mean-field — accounts for the fact that the *infector* of a newly infected node is itself one of its n neighbours, so only $n-1$ neighbours are at risk, *and* one of those is excluded by the pair-approximation excess-degree correction $\kappa = (n-1)/n$. The factor reduces to $n-2$ when the leading order of the next-generation matrix is computed.

Clustering reduces R_0 further; the package implements approximate corrections for Keeling and Barnard closures.

Setup

```
using NodeBasedModels
using Plots
```

Numerical R_0

```
net = regular_network(6)
R0_b = basic_reproduction_number(net, BernoulliClosure(), 0.2, 0.1)
println("Bernoulli, n=6,  $\tau=0.2$ ,  $\gamma=0.1 \rightarrow R_0 =$ ", R0_b)

net_cl = regular_network(6;  $\phi = 0.4$ )
R0_k = basic_reproduction_number(net_cl, KeelingClosure(), 0.2, 0.1)
println("Keeling, n=6,  $\phi=0.4 \rightarrow R_0 =$ ", round(R0_k; digits=3))
```

Bernoulli, $n=6$, $\tau=0.2$, $\gamma=0.1 \rightarrow R_0 = 8.0$

Keeling, $n=6$, $\phi=0.4 \rightarrow R_0 = 6.72$

Symbolic R_0

```
R0_sym = basic_reproduction_number(sir_model(), net, BernoulliClosure())
println("Symbolic  $R_0 =$ ", R0_sym)
```

Symbolic $R_0 = (4\tau) / \gamma$

Threshold sweep

We sweep τ and record the final epidemic size:

```
function final_R( $\tau_{val}$ ;  $\gamma_{val} = 0.1$ ,  $n = 6$ ,  $T = 400.0$ )
    net = regular_network(n)
    psys = generate_pairwise(sir_model(), net, BernoulliClosure();
                            $tspan = (0.0, T)$ ,  $N = 1.0$ ,  $seed\_fraction = 0.001$ )
    p = copy(psys.params);  $p[:\tau] = \tau_{val}$ ;  $p[:\gamma] = \gamma_{val}$ 
    sol = solve_pairwise(psys, p;  $reltol = 1e-8$ ,  $abstol = 1e-10$ )
```

```

    return sol[psys.singles[:R]][end]
end

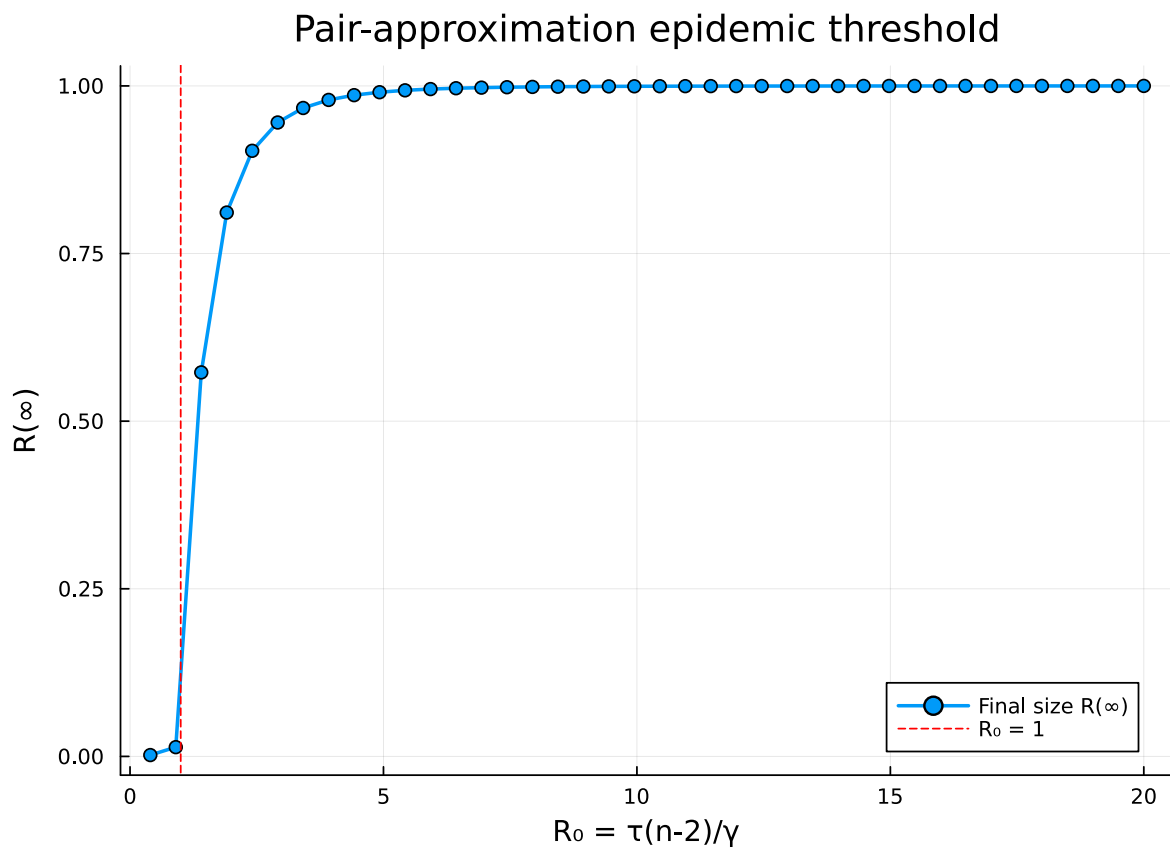
τgrid = range(0.01, 0.5; length=40)
R∞ = [final_R(τ) for τ in τgrid]
R0 = [τ * 4 / 0.1 for τ in τgrid] # n=6 → n-2 = 4
nothing

```

```

plot(R0, R∞, lw = 2, marker = :circle, label = "Final size R(∞)")
vline!([1.0], ls = :dash, color = :red, label = "R₀ = 1")
xlabel!("R₀ = τ(n-2)/γ"); ylabel!("R(∞)")
title!("Pair-approximation epidemic threshold")

```



The threshold at $R_0 = 1$ is sharp in the deterministic pair-system: subcritical seeds die out, super-critical ones grow.

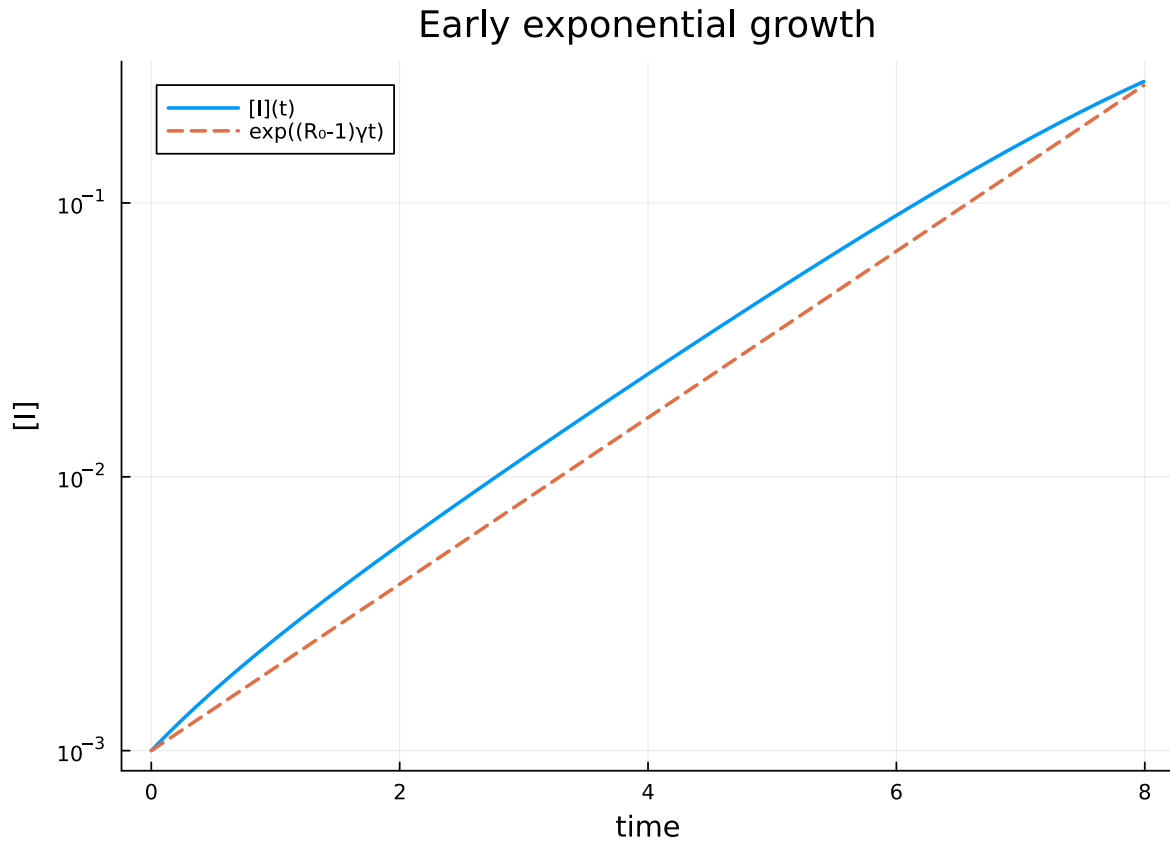
Early growth

For supercritical $R_0 > 1$, $[I](t)$ grows exponentially with rate $r = (R_0 - 1)\gamma$ at early times. Let us verify:

```
net = regular_network(6)
psys = generate_pairwise(sir_model(), net, BernoulliClosure();
                        tspan = (0.0, 30.0), N = 1.0, seed_fraction = 0.001)
I0 = 0.001
p = copy(psys.params); p[:τ] = 0.2; p[:γ] = 0.1
sol = solve_pairwise(psys, p; reltol = 1e-10, abstol = 1e-12)

t_early = sol.t[sol.t .< 8]
I_early = sol[psys.singles[:I]][1:length(t_early)]

plot(t_early, I_early, yscale = :log10, lw = 2, label = "[I](t)")
plot!(t_early, I0 .* exp((8.0 - 1.0) * 0.1 .* t_early),
      lw = 2, ls = :dash, label = "exp((R0-1)γt)")
xlabel!("time"); ylabel!("[I]")
title!("Early exponential growth")
```



Summary

`basic_reproduction_number` returns numeric or symbolic R_0 for any model–network–closure triple; the threshold $R_0 = 1$ separates die-out from outbreak in the pair system.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use [NetworkOutbreaks.jl](#)'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette [01_sir_on_graphs](#) for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.